

Group Report: Part 1 – Regression and Classification Activity

Members: Ahammad Jony, Kumara Muditha, Podkorytov Nikolai, Shrestha Bata Prabesh

Domain: Smart Cities & Sustainable Transportation

Repository and Commit

- **Repository:** <https://github.com/Muditha-Kumara/Applied-artificial-intelligence>
- **Branch:** `main`
- **Commit:** `6ba27dc2f41c60120bb10bd25e2209f0788f71e3`

1. Problem Statement Revisited

The purpose of this part of the project is to understand how Helsinki might deal with the increasing number of electric vehicles by 2030. We focused on estimating how much charging demand different districts may generate and what kind of charging infrastructure would be needed.

Regression task: estimate the total power demand (kW) for each district in 2030.

Classification task: decide whether a district should be equipped mainly with standard Type 2 chargers or if it will need fast-charging stations.

2. Dataset Description and Analysis

We worked with three datasets:

1. **Helsinki Population Forecast (2018–2030):** Provides historical growth and future projections.
2. **Helsinki EV Charging Infrastructure:** Current registry of 240+ stations used to calculate existing supply and power bands.
3. **Vehicle Registry (Helsinki Area):** Analyzed to understand the correlation between vehicle mass/power and fuel types.

One thing that stands out is that most current chargers are 22 kW AC. Considering the expected growth of EVs, some districts will clearly need more powerful DC chargers (above 50 kW), especially those that are growing quickly.

3. Data Splitting Process and Rationale

After cleaning the data and adding new features such as population growth rate, we split the dataset into 80% for training and 20% for testing.

For the classification task, we used a stratified split so that the few high-growth districts would not disappear from the training set.

The test set helped us check whether the models behave reasonably on districts they have not seen before.

4. Regression Results

For the regression part, We set up a small pipeline with scaling and a basic linear regression model. The idea was simply to see whether population numbers and their change over time could explain the expected charging demand. The features we used were the population in 2022, the projected population for 2030, and how much the district grew between 2018 and 2022.

The model ended up with an R^2 score of 0.85. It's not perfect, but it shows that these population-based features already capture most of what drives the demand. We also checked the residuals just to make sure the model wasn't consistently over- or under-predicting for certain districts. Nothing unusual showed up there, so the model behaves more or less evenly across the dataset.

5. Classification Results

We compared two models: a Decision Tree and Logistic Regression.

Model	Accuracy	Key Observation
Decision Tree	91%	Excellent at capturing non-linear "tipping points" for infrastructure hubs.
Logistic Regression	86%	Provided a stable linear baseline but was less sensitive to rapid growth outliers.

Overall, the Decision Tree turned out to be more suitable for identifying districts that will likely need fast-charging hubs.

6. Ethical Risk Assessment and Mitigation

- Risk 1: Spatial Bias (Severity: High).** If decisions are based only on population growth, some low-growth or low-income districts might be overlooked.
Mitigation: planners should review the model’s suggestions and ensure that every district gets at least a basic level of charging access.
- Risk 2: Data Obsolescence (Severity: Medium).** Charging technology and EV usage change quickly, so the assumptions used today may not hold in a few years.
Mitigation: retrain the models regularly using updated EV registration data.

7. Key Insights and Conclusions

- Growth Velocity Matters:** Simply looking at total population is insufficient; the *speed* of growth is a critical feature for predicting where infrastructure will be overwhelmed first.
- Infrastructure Gap:** There is a clear need to transition from residential-style AC charging to high-capacity DC hubs in the Southeastern and Central

districts.

3. **Responsible AI:** By combining automated predictions with ethical oversight, Helsinki can achieve an equitable EV transition that supports climate goals without leaving smaller districts behind.

Submission Attachments:

1. **Python Notebook (2.1 part1.ipynb):** Contains the complete pipeline: data cleaning, feature engineering (Growth Rate), Linear Regression with Scaling, and Decision Tree/Logistic Regression comparisons.
2. **Visualizations:** *Regression Fit: Population vs. Power Demand.*
 - *Residual Analysis:* Diagnostic plot for regression.
 - *Confusion Matrices:* Performance comparison for classification tasks.



